

vision**2**learn

Cyber Security Level 3

Hints
and
Tips

Unit 1



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Hints and Tips Unit 1

The information in this guide will be helpful to you when you are ready to complete your assessment:

Unit 1: Understand cyber security principles Level 3

Section 1 – Understand cyber security

Q1. **Describe** your understanding of the concepts of cyber security in the box below. You will need to include the following in your answer:

- Security
- Identity
- Confidentiality
- Integrity
- Availability
- Threat
- Vulnerability
- Risk
- hazard

This is a describe question and will therefore need a detailed answer. You should include reference to each factor in the bullet point list within the question.

Q2. **Explain** why cyber security is important

This an explain question and will therefore need a detailed answer. You should state why cyber security is important and then explain the reasons for why it's important.

Q3. In the box below, **Describe** the consequences and implications for unauthorised access to the distribution, or loss, of the following. Include each of the following in your answer.

- sensitive data
- personally identifiable information (PII)
- protected health information (PHI)
- personal information
- intellectual property
- industry information systems

This will need a detailed answer where you include reference to each bullet pointed factor and include the consequences and implications for each. These must relate to unauthorised access to the distribution, or loss, of each factor.

Section 2 – Understand core terminology and key aspects of cyber security

Q1. In your own words, **Define** each of the following:

- Malicious software
- Distributed denial of service (DDos)
- Cloud
- Software
- Domain
- Exploit
- Breach
- Firewall
- Encryption
- Bring Your Own Device (BYOD)
- Penetration testing (pen testing)

Using the table provided, provide a definition of each term. You must ensure that this is not copied from those in the course material. You may include an example to help you demonstrate your understanding in your own words.

Important: When talking about domains you need to consider network domains in your response. Please see the following website for more information on this.

[What is a Windows Domain and How Does It Affect My PC? \(howtogeek.com\)](https://www.howtogeek.com/101446/what-is-a-windows-domain-and-how-does-it-affect-my-pc/)

Q2. Complete the following for good actors and bad actors:

a) **Explain** each term:

Your explanation for each type of actor should distinguish the different between the two. You may include the types of people that might be each type of actor to help with your explanation.

b) What behaviours do each exhibit? Distinguish the difference between each:

In each box, describe the different behaviours you might expect from each type of actor. You will need to ensure that these are different for each type of actor. You may include examples for what each might do to help with your explanation.

c) Explain the motivations of each:

Think about why the different types of hacker might do what they do, what will they gain? Can one become the other and why might this happen?

Q3. Identify five key sectors that are most vulnerable to a cyber-attack

As this is an identify question, you may list the sectors. Include at least 5 in the table. You may add more if you wish.

Q4. Thinking about the key sectors above:

a) Select two of the key sectors above and compare the motivations for a cyber-attack

You may choose any two key sectors. You must compare the motivations for why they may have a cyber attack. To compare you must describe the similarities and differences for why attacks might happen.

b) For any one of the key sectors, consider how an actor may carry out a cyber-attack

For this answer, think carefully about how an attack might be carried out on one of the key sectors. Make your answer specific, include the type of attack and what steps the actor might take.

Section 3 - Understand security by design principles

Q1. Describe the term security by design This should be a detailed description.

You may include an example to help fully demonstrate your understanding.

Q2. Read the above case study, then in the box below, write a report to include:

- Why PaperX should consider security by design
- Which principles of security of design are best suited to PaperX and why
- The consequences for PaperX of not considering cyber security during the design phase
- An evaluation of the advantages and disadvantages of security by design

This report will cover four learning outcomes and will therefore be very comprehensive. You may use any layout that suits you, as long as it's professional. You may attach the report as a separate document if you prefer.

Before submitting your report, check that it does cover each bullet point. Does it include your own thoughts as to why your design is the best for this scenario? Have you fully explored the advantages and disadvantages of your design? Is there another design that could be better for some parts?